University of Pisa

Department of Information Engineering

Master Degree in Artificial Intelligence and Data Engineering

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GeoRisk

Project Documentation

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Contents

1. Problem/Solution Fit

Italian insurance companies and loss adjusters today assess the landslide and flood risk of a specific building with hazard maps that are too coarse, because they refer to the whole municipality. GeoRisk is a platform that takes the address of a building, combines Italian public geographic data with an artificial intelligence model, and returns in a few seconds a risk score for that building, not for the whole area.

1.1 Information Sources and Methods Used

1.1.1 Academic Research

There are three strands: the feasibility of AI scoring at building level, the limits of zonal mapping, and the Italian market context.

- **FloodGenome** (Liu and Mostafavi, 2024) [1] and **Dal Sasso et al.** (2026) [2]: granular scoring on geospatial features for flood risk is an active research area, and it can be applied in practice (taken up again in Section 1.3.2).
- **Perazzini, Gnecco and Pammolli** (2020) [3]: they model the pricing failure that comes from insufficient granularity in Italian cat-nat insurance (taken up again in Section 1.2).
- **Bellaver et al.** (2025) [4]: the first systematic study on the under-pricing of flood risk in the Italian property market when updated information is missing (taken up again in Section 1.2).

1.1.2 Patent Research and Freedom to Operate

On Google Patents and USPTO we found four relevant patents in geospatial scoring of insurance risk: US 12,205,184 B2 [5], US 11,294,552 B2 [6], US 11,669,794 B2 [7], all filed in the United States, and EP 4,014,179 A4 [8], the only one also filed in Europe.

Nothing blocks us. Patents are territorial, so the US families have no effect in Italy. Our system also stays outside the claims of all four. The closest one, US 12,205,184 B2, claims only single-factor scoring (inside or outside a zone), the limit we criticise in Section 1.3.2, and not a two-branch model trained on historical events. EP 4,014,179 A4, the only one also filed in a European jurisdiction, was later withdrawn, so it is not an active patent. Even if we consider it for completeness, it covered a generic pattern matching between risk scenarios, not the combination of Italian geo-hydrological data with a double landslide and flood scoring. No patent covers our use case: probabilistic dual-peril scoring, based on Italian historical events, explainable through SHAP.

1.1.3 Regulatory and Institutional Sources

Three regulatory developments make the problem urgent, and not only real.

Since 2024 Italian businesses must insure by law against catastrophic events such as landslides and floods (L. 213/2023) [9]. In September 2026 this duty is fully active for almost all categories. However, according to ANIA only about 15% of the eligible businesses have an active policy, out of a potential of 2.4 billion euro of yearly premiums, of which 700 million are collected [10]: 1.7 billion are uncovered.

The revised Solvency II Directive (EU 2025/2), to be transposed by January 2027, requires companies to include climate risk scenarios in their internal risk assessment [11], but it also introduces a "simplified" category for smaller companies [12]. The regulator itself recognises that these tools are today out of their reach.

The AI Act classifies AI pricing systems as "high risk" only for life and health, and not explicitly for non-life insurance [13]. That counts in favour of GeoRisk, although lawyers still debate where the boundaries lie (discussed further in Section 1.5 and in Chapter 6).

1.2 Problem Definition

The ISPRA report on hydrogeological instability (2024 edition) states that 94.5% of Italian municipalities (7,463 out of 7,896) fall in areas exposed to landslides, floods, coastal erosion or avalanches; 1.28 million people live in areas with high or very high landslide hazard (P3-P4), and 6.8 million in areas with medium hydraulic hazard; 69,500 km^2 are at landslide risk, which is 23% of the national territory [14]. Geo-hydraulic risk is the territorial norm, not the exception. Still, only about 15% of businesses have adequate cover despite the legal duty (Section 1.1.3). This gap reflects a structural problem in the way risk is assessed and priced.

The people who experience it are the independent broker or loss adjuster, and the underwriting team of a small or mid-sized insurer, who must decide whether to insure a building and at which premium. Today they check the official hazard maps (PAI) by hand. These maps are binary and have zonal granularity: they say whether an area falls in a risk band, but they do not tell apart a building 50 metres from a watercourse and one 3 km away in the same administrative band. The process is slow, it needs fragmented sources (regional portals, land registry, event history) and it does not scale to the volume imposed by the 2024 cat-nat duty.

Insurers pay the cost: undifferentiated prices for very different exposures in the same area, with over-pricing for safe customers, who therefore do not buy voluntarily, and under-pricing for exposed properties, whose losses are not covered by adequate premiums, exactly the mechanism described by Perazzini, Gnecco and Pammolli (Section 1.1.1, [3]).

Climate risk scoring tools for single buildings already exist, but they are built for the enterprise market (large international companies, mostly from the US and the UK), with costs and integration complexity out of reach for an Italian regional insurer or a small firm of loss adjusters. The Italian public maps (PAI/ISPRA) are authoritative and free, but they remain a visual reference for land planning, not a score that can be integrated into underwriting. There is indirect proof of the gap: the CINEAS white paper (Politecnico di Milano, 2023) reports that about 75% of Italian homes are exposed to "significant" risk and less than 5% have adequate cover [15], the same gap seen from the side of the housing stock.

1.3 AI-Enabled Solution

1.3.1 End-to-End System Overview

GeoRisk is a B2B platform. Given the address of a building, it returns in a few seconds an overall score (0–100) for two geo-hydrological hazards: landslide and flood. The operational flow (Section 3.2) is fully automatic. Starting from the address entered by the operator (broker, loss adjuster, underwriter), the system runs the geocoding, collects in parallel six categories of public geospatial data (building geometry, hydrogeological hazard, hydrographic data, ground elevation and historical events for both risks), computes the features for each branch and returns a combined score with an explanation. No step needs manual work. This, more than data availability, is what makes GeoRisk different from the loss adjuster, who checks the same sources one at a time.

1.3.2 Why AI Is Needed and GIS Is Not Enough

A PAI map answers a binary and static question: "does this point fall inside or outside a risk band?". Every manual assessment made today starts from it, but it is not enough for accurate pricing. **First**, zonal classification treats in the same way buildings with very different physical exposure inside the same band, and as shown in Section 1.1.1 this is what produces the mispricing described in Section 1.2. **Second**, a hazard band is not updated at the speed at which real risk patterns change: a building just outside the official perimeter, but close to points that flood again and again, is not told apart from a truly safe one. **Third**, a binary threshold does not allow us to weigh several correlated factors at the same time (distance from the watercourse, elevation, historical frequency of events) in one gradual score. FloodGenome and the framework of Dal Sasso et al. (Section 1.1.1, [1][2]) show that models trained on multiple hydrological, topographic and historical features produce finer classifications than a zonal map, and remain interpretable.

A model trained on where past events really happened learns the relative weight of the factors from the data itself, instead of from a threshold fixed in advance, and that qualitative jump is what justifies AI over a deterministic GIS.

1.3.3 Model Architecture

Landslides and floods have partly different physical drivers. For landslides these are slope and absolute elevation (TINITALY digital elevation model by INGV [16]). For floods they are the distance from the nearest watercourse (river network from the `waterway` tag of the OpenStreetMap Overpass API [17]) and the elevation relative to the surrounding area. Above all, structured historical data have a different availability in Italy for the two hazards. For this reason GeoRisk uses two separate sub-models, combined into one final score.

Landslide branch: supervised probabilistic classification. The target comes from crossing the position of the building with the Italian Landslide Inventory (IFFI [18], ISPRA/IdroGEO) at several metric radius thresholds (30 m, 500 m, 2 km), as IdroGEO already does in its own interface. **Flood branch**: same logic, with the target taken from the historical events of the international EM-DAT database [19]. Since only part of the records report precise coordinates, the match is hybrid: a metric radius where coordinates exist, and the province as a fallback. The official hydraulic hazard (PAI) enters both branches as an input feature, not as a target.

Granularity is therefore openly asymmetric between the two branches: not in the structure, which is identical, but in the quality of the historical signal. We take up this real limit again in Section 6.2.4. The count of historical events that builds the target is computed within a reference time window. This window is an explicit design parameter and is not derived from the datasets, which set no time limit: it balances a wide sample of positive events with a signal that represents current risk.

The two sub-scores are combined by taking the **maximum**. The criterion is prudential: a building with high risk for even one hazard type stays classified as such, and an average does not dilute a risk concentrated on a single factor.

A cross-cutting requirement is **explainability**. The reformed Solvency II Directive

(Section 1.1.3) asks for transparency in risk management models and for explicit supervision of the calculation systems, and the transparency framework of the AI Act (Section 1.1.3) goes in the same direction. The underwriter must be able to justify the score given to a property in order to meet compliance standards. The candidate model therefore belongs to boosted decision trees (Gradient Boosting and Random Forest), which break down each prediction into the contributions of the single factors through SHAP (SHapley Additive exPlanations) [20].

1.3.4 Output

For each address the operator receives: the final overall score (0–100), which is the maximum of the two sub-scores, with a colour band that is easy to read at a glance; the SHAP breakdown of the factors of the branch that determined the score; a confidence indicator, separate for each branch, that flags when the historical coverage of the data is poor, which matters in particular for the flood branch given the asymmetry of Section 1.3.3; and an exportable report for the internal documentation of the file. The score expresses the estimated probability of an event within the reference time horizon used in training.

1.4 Achieving Product-Market Fit

GeoRisk addresses two B2B segments already introduced in Section 1.2: the **independent broker or loss adjuster**, a wide and fragmented population of mostly small and local firms; and the **underwriting team of a small or mid-sized insurer**, where, besides the person who assesses and uses the tool, there is typically a compliance role that authorises its adoption (discussed further in Section 5.1).

The fit does not come from a generic need for digitalisation. It comes from the match between the technical features of the system (accuracy at single building level, immediate answer, output explainable through SHAP) and the constraints of the two segments: a limited budget for enterprise tools, a volume of assessments that grew because of the cat-nat duty, and the need to document the risk methodology for Solvency II purposes. Consistently with this, the launch is not simultaneous across the whole country. We start from a pilot area, **Tuscany**, to validate the product with the first customers before facing the cost of expansion to the remaining Italian hydrographic areas (Chapter 3).

1.5 Logical Assumptions Foundation

- **Willingness to pay for greater accuracy**: brokers, loss adjusters and small and mid-sized insurers see the zonal granularity of PAI maps as a concrete limit to their work, and they are willing to pay for a tool that solves it, not only to use it if it is free.
- **Public historical data are adequate as a proxy of real risk**: without access to the claims that were actually paid, which is confidential commercial information, the model trains on historical events from public sources (IFFI, EM-DAT). We assume this is an accurate enough approximation, while we recognise that a recorded event does not necessarily match a paid claim.
- **Asymmetric granularity between the branches is sufficient**: we assume that a combined score with separate confidence indicators for each branch (Section 1.3.4) is enough to manage the difference in reliability between the landslide branch, validated on point events, and the flood branch, which has coarser granularity, without compromising practical usefulness for the underwriter.

- **AI Act compliance**: we assume that the product does not fall among “high risk” AI systems, because the law states this explicitly only for life and health, and not for non-life insurance.
- **Stable access to automated data sources**: we assume that the ISPRA WFS, the Overpass API and EM-DAT stay accessible in a programmatic and stable way, without commercial restrictions or interruptions that would compromise the end-to-end automation of Section 1.3.1.
- **The basic mechanism is not defensible in itself**: we assume, consistently with Chapter 2, that anyone with access to the same public sources can replicate the technical component. Defensibility lies elsewhere, in the relationship with the first customer and in the proprietary dataset built over time, and not in the architecture.
- **Width of the historical time window**: we assume that a reference window, roughly 15–30 years and to be defined empirically during validation, is an adequate trade-off between the number of positive events in the sample and how well they represent current risk.

2. Competitive Analysis

2.1 Competitive Landscape

2.1.1 Direct competitors

- **Verisk WaterLine** [21]: a flood risk score from 0 to 100 for each single property, built from probabilistic hydraulic simulations. The concept is the same as GeoRisk. However, it covers only the US, it needs enterprise licences and it does not deal with landslide risk.
- **Cytora**, with the GeoSmart Information partnership [22]: flood risk data for each single property inside its own risk processing platform for insurers, but as an enterprise product for the UK and global market, with no Italian coverage and no PAI data.
- **SaferPlaces** [23]: an Italian flood intelligence startup based on AI and satellite observation, already used in the field by the Civil Protection of Emilia-Romagna during the 2024 floods. In geographic terms it is the closest competitor, but it was born for emergency mapping of the territory, not for underwriting, and it does not deal with landslide risk.
- **Previsico** [24] (UK): probabilistic flood forecasting, with Generali among its customers. This proves that Italian insurers already buy solutions of this type. It remains an enterprise supplier for large companies, it works only on floods, and it is not designed for independent brokers or regional insurers.

2.1.2 The closest prior-art case

The most relevant case is a proof-of-concept officially documented by EUSPA, the EU Agency for the Space Programme [25]. Vittoria Assicurazioni and the Italian startup Eoliann developed a flood risk index (0–10) at single asset level, based on Copernicus satellite data and AI. The work was explicitly motivated by the 2024 Budget Law, the same regulation discussed in Chapter 1. The PoC showed a measurable improvement over the previous GIS solution: correct identification of real events went from 67% to 78%, and resolution went from 200 m to 30 m. This led to a structural commercial partnership.

The PoC is a real Italian precedent that the jump from a static GIS to an AI-driven system produces a concrete advantage, which is the argument of Section 1.3.2. But Eoliann covers only floods, not landslides, and it is built for large companies like Vittoria. It does not offer an accessible path for independent brokers or small regional insurers, the segment where GeoRisk positions itself.

2.1.3 Indirect competitors and substitutes

The **PAI/ISPRA maps** consulted by hand meet the same basic information need, but without scoring, without automation and with zonal granularity, the limit described in Chapter 1. The second element is the large data aggregators rooted in the Italian financial market, in particular CRIF [26] and Cerved. Both are active in commercial information services and Automated Valuation Models for Italian banks and insurers. Neither of them does climate risk scoring yet, but both already have the commercial relationships needed to extend in this direction, a plausible entry that we take up again in Chapter 6. The most common substitute, however, remains the process used today by independent brokers and loss adjusters: a manual assessment by eye, with no premium differentiation inside the same risk zone.

2.2 Assessing Competitors

The pattern is clear along three axes: peril coverage, segment served and presence in Italy. The **international enterprise players** (Verisk, Cytora) bring technological strength and large capital, but their strength is also the reason why they stay out of reach for the Italian SME/broker segment: their infrastructure is mature, and it is built for large companies with a dedicated IT budget. The **Italian players** (Eoliann, SaferPlaces) have the opposite advantage, a proven local relevance. Their limit is breadth, not quality: none of them deals with landslide risk, and none of them is designed for a customer other than the large insurer (Eoliann and Vittoria) or the public body (SaferPlaces and the Civil Protection). The **PAI/ISPRA maps** have institutional authority and zero cost, but they remain a lookup and not a score, which is exactly why they do not solve the granularity problem of Chapter 1. CRIF and Cerved have the most dangerous advantage: widespread commercial relationships with Italian banks and insurers, but neither of them has chosen to enter this space yet.

Competitor	**Peril coverage**	**Target customer**	**Presence in Italy**
Verisk WaterLine, Cytora	Flood only	Enterprise	No
Eoliann, SaferPlaces	Flood only	Large insurers / public bodies	Yes
Previsico	Flood only	Enterprise	Italian customer (Generali), not a local product
PAI/ISPRA	Landslide + flood	Anyone, but manual	Yes (a source, not a product)
CRIF, Cerved	None (potential)	Banks and insurers	Yes
GeoRisk	**Landslide + flood**	**SMEs and independent brokers**	**Yes**

Table 2.1: Competitors compared by peril coverage, target customer and presence in Italy.

No competitor occupies the same cell on all three axes. GeoRisk positions itself in that gap, and we develop the point in Section 2.3.

2.3 Strategic Positioning

We summarise the positioning with the **ERRC** grid (Eliminate-Reduce-Raise-Create) [27] in Figure 2.1, which shows what the product eliminates and reduces compared to the enterprise competitors, and what it raises and creates from scratch for the segment that is not served yet. We then translate it into a **Strategy Canvas** [27] on six axes, shown in Figure 2.2, which compares GeoRisk with the three competitive clusters identified in Sections 2.1 and 2.2.

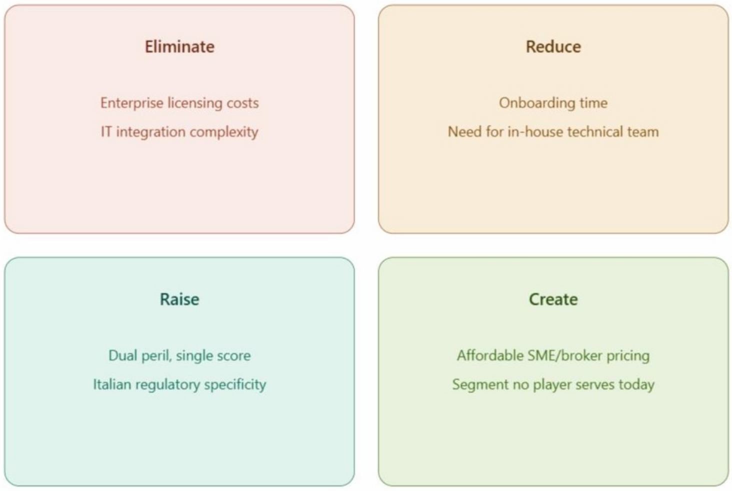

Figure 2.1: ERRC grid of GeoRisk's positioning with respect to enterprise players, Italian players and PAI/ISPRA maps.

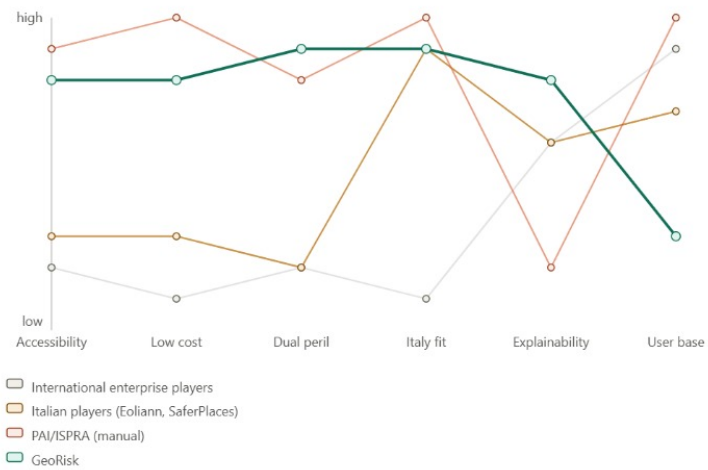

Figure 2.2: Strategy canvas: comparison between GeoRisk and the three competitive clusters along six positioning axes.

GeoRisk does not come first on every axis. The PAI/ISPRA maps stay cheaper, more widespread and more rooted in the sector. But GeoRisk is the only one that combines dual-hazard coverage and analytical explainability, and those are the two dimensions an underwriter needs in order to use public data for pricing.

2.4 Validating Competitiveness

We are not entering an empty market. The basic mechanism, a spatial join on public data plus gradient boosting, is not defensible in itself (Section 1.5), and the Eoliann and Vittoria case (Section 2.1.2) shows that we do not even claim it is original. That remains our real weak point. The question is not "is the idea new?" but "can we really compete?".

First, no player occupies the position of GeoRisk today. The Strategy Canvas (Section 2.3) shows that no cluster combines dual hazard coverage and analytical explainability with the accessibility that the SME/broker segment needs.

Second, the position is not permanently safe. CRIF or Cerved could add a Cat-Nat module by using their existing commercial relationships (Section 6.2.3). Real defensibility works on two levels. The primary barrier is the proprietary dataset built over time from real claims collected from customers, which grows stronger only with use. The secondary barrier is the specialisation on the dual peril and on the Italian regulatory context, which is easier to imitate and therefore less solid. Neither of the two exists at launch: the first year is a window of maximum exposure, where we are visible to competitors but not yet defended, and it is described as a failure cause in Section 6.2.3. If in that window we do not win a first reference customer before a player like CRIF replicates the offer, the positioning advantage runs out without ever becoming a barrier.

Third, the advantage lies where the competitors are weak: free but not automated for the PAI/ISPRA maps, automated but out of reach for the enterprise players, Italian but single-peril for Eoliann and SaferPlaces. None of them covers the three things together.

2.5 Industry Forces

We close the chapter with Porter's five forces [38], plus the sixth force of complementors. The aim is not to classify the sector, but to see which pressures the plan has to answer.

Threat of new entrants: high. The mechanism is a spatial join on public data plus a gradient boosting, and Section 1.5 already admits it is not defensible: anyone with the same sources can rebuild it. What keeps the field thin is the data engineering cost of Section 3.4, which slows us down as much as anyone else.

Bargaining power of suppliers: high. Our inputs come from ISPRA, INGV, the Basin Authorities, OpenStreetMap and EM-DAT, and Section 3.1 states that none of them has an agreement with us. They can cap or stop a service without warning, and Section 6.2.6 describes what happens then. The commercial geocoder adds a second dependency, since the free Nominatim instance cannot be used in production (Section 3.2.1). The PostGIS mirror is our answer to this force.

Threat of substitutes: high. The PAI portals are free, a manual assessment by eye costs nothing, and the loss adjuster often gets a free rough estimate from the mandating insurer (Section 6.2.2). We compete against a free habit, not a paid product. That is why the plan starts with a free trial and not a discount.

Bargaining power of buyers: mixed, and it moves. A small firm can leave at any moment, so its power stays high, and the churn assumptions of Section 4.2.3 reflect that. An insurer is slow to buy but hard to replace once the API sits inside its underwriting process, which justifies the four-year retention of Section 4.3.2.

Rivalry: low today, and only today. Table 2.1 shows that no player covers dual peril, Italian data and an accessible price at once. But the demand side is a fragmented industry with no pricing power, entry barriers are low and we own almost no fixed assets, so rivalry can rise as soon as the niche looks worth taking.

Complementors: the force to watch. The software houses of agency management systems are complements today, because their product carries ours onto the desk of the broker. They turn into the threat of Section 6.2.3 the moment they bundle a Cat-Nat module of their own.

Three of these forces are clearly against us, one is mixed and rivalry is mild only for now. We read this as a reason for the plan, not an argument against it. In an industry this exposed the Layer 3 archive is the only asset worth building, and it has to be built before the window described above closes.

3. Business Model Canvas

Chapter 1 established that the problem is real and that the AI solution makes technical sense, and Chapter 2 placed GeoRisk among the existing players. Here we describe the mechanism through which GeoRisk creates value, how it delivers it and how it keeps part of it as revenue. The prices and costs set here are the input of Chapter 4.

GeoRisk is a B2B platform in SaaS mode, available through a web interface and a REST API. In the initial configuration it covers landslide and flood on the Tuscany pilot area, with the goal of national expansion (Section 3.4). It does not sell policies and it does not replace the judgement of the professional. We adopt the Business Model Canvas of Osterwalder and Pigneur [28], shown in Figure 3.1. Two blocks weigh more than the others: key resources, which are almost entirely data and models, and the cost structure. We look at them in more detail in Section 3.2.

3.1 Core Components Overview

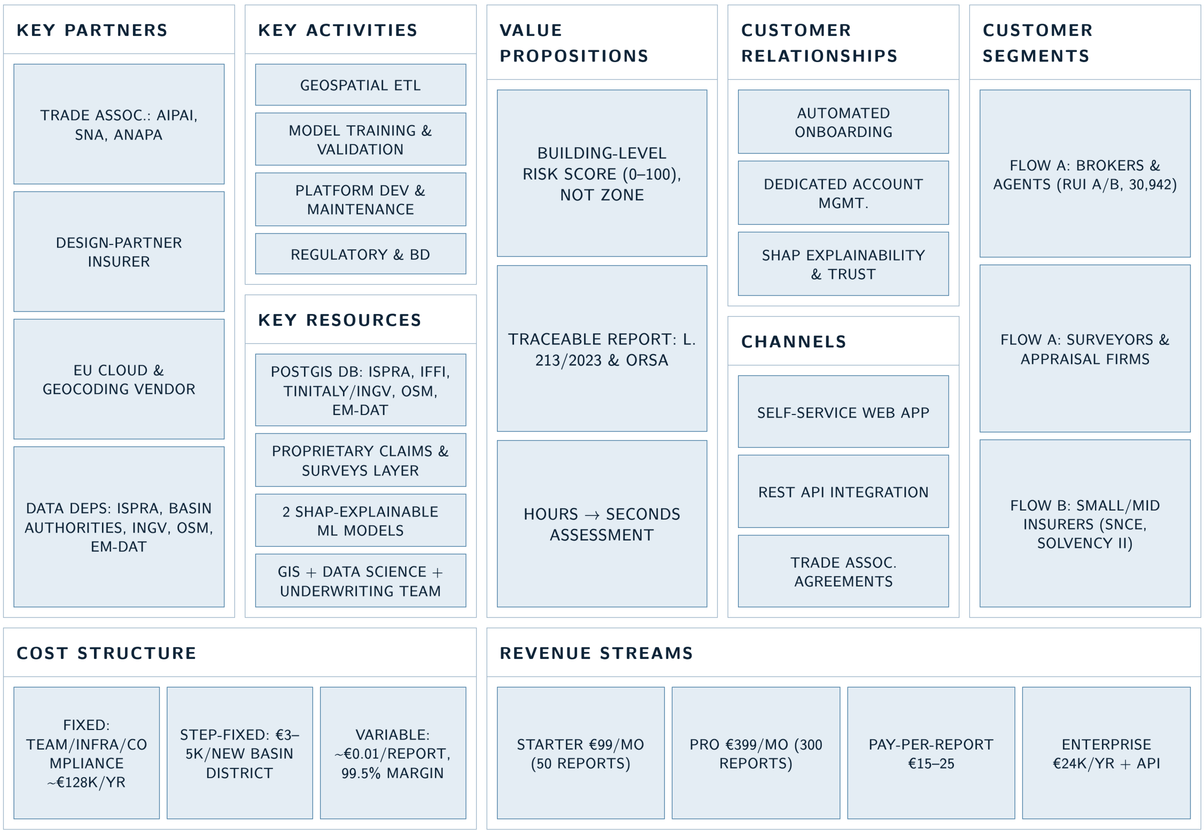

Figure 3.1: Business Model Canvas.

Customer Segments, already identified in Section 1.4 and described as personas in Section 5.1, are two. *Intermediaries*: a loss adjuster or independent broker listed in the RUI, a small local firm, who buys a SaaS subscription and decides on their own. *Small and mid-sized insurers*: a non-life insurer with SNCE status and a regional property portfolio, which buys an annual licence with API access, where Underwriting decides and Compliance authorises.

The order reflects the sequence of market entry, not economic importance. Intermediaries come first because we reach them without a sales force, in weeks instead of months, and because they generate the proprietary data of Section 3.2.2. Insurers have a long cycle but a unit value one order of magnitude higher, and they determine the sustainability of the business. Large national and international companies stay out: they already have enterprise catastrophe models (Chapter 2) and lack the budget constraint that defines our segment.

Value Propositions, based on four technical features from Chapter 1:

- *Granularity at building level*: a score on the geometry of the single building, not on the municipal band (Section 1.3.2).
- *End-to-end automation*: the six sources of Section 3.2.1 queried in parallel in a few seconds. The value is not access to the data, which is public, but the removal of the manual collection work.
- *Explainability as a commercial requirement*: the SHAP breakdown (Section 1.3.3) serves the underwriter, who must justify a premium, and Compliance, which documents the methodology for the ORSA. It is what makes the product purchasable by a supervised insurer, and not merely useful.
- *Affordable price*: about one order of magnitude below enterprise catastrophe modelling solutions.

One point about wording, which is also a legal defence. GeoRisk produces reports with a traceable methodology, not "certified reports": no third party certifies them. And it provides documentation that supports the obligations of L. 213/2023, but it does not "fulfil the cat-nat duty", which stays with the business and is fulfilled by taking out a policy. The distinction matters: the civil liability scenario of Section 6.2.4 is defended with liability exclusion clauses, which under Italian law do not cover wilful misconduct and gross negligence (art. 1229 of the Italian Civil Code), and saying "certified" would make the reliance of the broker legitimate and the clause ineffective.

Channels and Customer Relationships follow from the asymmetry between the segments. Italian adjusting firms are thousands, small and scattered, so selling to them one by one would cost more than a fee of a few hundred euro per month brings in: the channel is **associations** (AIPAI [29], about 300 firms and more than 2,000 specialised loss adjusters, plus SNA, ANAPA and trade fairs), with self-service registration, a free trial and ticket support. Insurers need the opposite, **direct selling** to the technical departments, the only workable channel when the purchase goes through Underwriting and Compliance, with assisted onboarding and support for the ORSA documentation, which is a substantial part of what they buy. The **design partner** is a case of its own: the first insurer or structured firm in the pilot area gets favourable terms in exchange for anonymised claims data, an exchange rather than a concession, discussed in Section 3.2.2 because it starts the proprietary dataset.

Revenue Streams (price list in Section 3.5) involve two structural choices. *The Enterprise price is not a volume discount*: at 10,000 assessments per year the €24,000 fee makes €2.40 per assessment, more than the €1.98 of the Starter plan. The insurer pays more for API access with service levels, for support and for the methodological documentation it uses with the supervisor. *Pay-per-report is not a fallback*: it defines the indifference point between occasional use and a subscription. At €99 per month against an average of €20 per report, the Starter plan is worth it only above five monthly assessments (€99 ÷ €20 ≈ 4.95). Below that, the rational choice is to stay on spot use. The same arithmetic generates the failed conversion scenario of Section 6.2.2.

Key Resources (detail in Section 3.2) are almost all intangible: the two trained models with the SHAP layer, the ingestion pipeline for the six geospatial flows, the PostGIS mirror that decouples the service from the real-time availability of the public APIs, the proprietary dataset of validated assessments, and the technical team with GIS and tabular machine learning skills.

Key Activities are the operation of the platform, the periodic retraining of the models, and regulatory monitoring on the AI Act, Solvency II, GDPR and dataset licences. The dominant one is **maintenance of the data connectors**, the most expensive and the least visible: every River Basin District Authority updates its layers with its own timing and schemas, and any change can break the ingestion.

Key Partners. Not every party we depend on is a partner. The **institutional data sources** (ISPRA, INGV, the seven River Basin District Authorities, OpenStreetMap through the Overpass API, CRED/UCLouvain for EM-DAT) are a one-way dependency: we have no bilateral agreement with any of them, and the issuing body can limit or suspend the service without notice, which is the failure scenario of Section 6.2.6. Calling them "partners" would hide that risk instead of exposing it. The real partners are the trade associations as a channel, the commercial geocoding supplier, the cloud provider and the design partner, the most important one in the early phase because it is the only source of the data the model does not get from public sources.

Cost Structure is almost entirely fixed: at about €0.01 of variable cost per assessment (Section 3.2.3) the whole economic problem moves onto fixed costs.

Item	**Nature**	**Note**
Technical team	Recurring fixed	The dominant item
Cloud infrastructure and PostGIS database	Recurring fixed	Grows slowly with volume
Data ingestion and validation for a new river basin district	One-off fixed, per territory	∼€3,000–5,000 per district
Maintenance of the existing connectors	Recurring fixed	Grows with the number of districts covered
Commercial geocoding	Variable	∼€0.005 per call
Legal and compliance advice	Recurring fixed	AI Act, Solvency II, GDPR, data licences
Marketing and association channels	Recurring fixed	–

Table 3.1: Cost structure of GeoRisk.

The relevant row is the third one: the cost does not scale with the number of assessments but with the territorial extension covered, the opposite of usage-based AI products, with consequences for scalability (Section 3.4).

3.2 AI-related Assets Focus

3.2.1 The End-to-End Pipeline

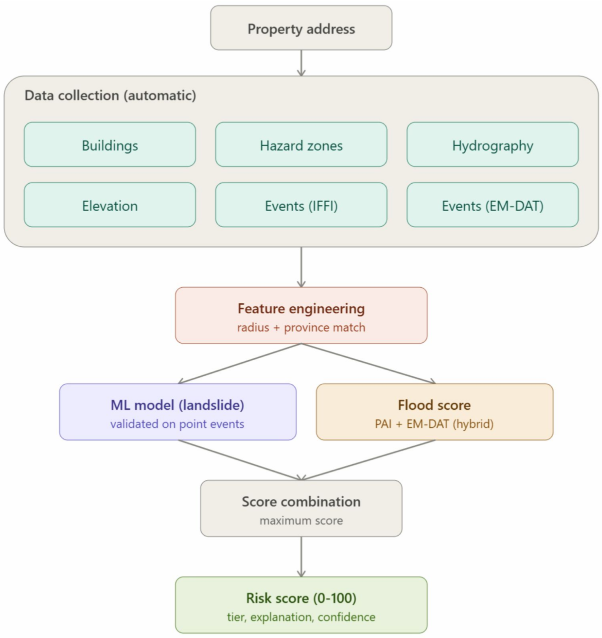

Figure 3.2: Processing pipeline.

Figure 3.2 shows the processing pipeline. Table 3.2 describes each phase.

Table 3.2: Phases of the GeoRisk processing pipeline.

Phase	Name	Technical description and tools
1	Input ingestion and geocoding	The user enters the street address. The system queries the geocoding service (Nominatim/OpenStreetMap) and gets WGS84 coordinates (latitude, longitude) and the province.
2	Metric projection (ETL)	Conversion into a Cartesian system in metres (UTM Zone 32N / EPSG:32632) through GeoPandas, for precise spatial computations (distances, buffers).
3	Data retrieval (parallel queries)	Parallel query of the six sources: building geometry (Overpass/OSM), landslide and flood hazard (IdroGEO/PAI), hydrography (Overpass, `waterway` tag), elevation (TINITALY/INGV), historical landslide events (IFFI), historical flood events (EM-DAT). Each query returns raw data (polygons, rasters, event lists), with no processing at this stage.

Phase	Name	Technical description and tools
4	Feature engineering	A vector of numerical and categorical features in three groups. **Common**: absolute and relative elevation (TINITALY), slope (from the DEM), distance from the nearest watercourse, landslide hazard band (P1-P4) and flood hazard band (Low/Medium/High), from the geometric intersection of the building × the PAI polygons. **Landslide branch**: count of IFFI events within multiple radii (30 m, 500 m, 2 km), type of the nearest landslide movement (for example fall, flow, slide, a categorical field in the IFFI records), year of the most recent event within the radius. **Flood branch**: a "province historically hit" flag (text comparison between the province of the building and the province field of the EM-DAT events), count of EM-DAT events within a metric radius where the coordinates are available (otherwise a fallback on the province flag), average severity of the matching events. All counts, for both branches, fall within a reference time window (roughly 15–30 years, Section 1.5).
5	Inference and risk scoring	The features of the landslide branch feed a first pre-trained model (Random Forest/XGBoost) that returns a probability of a landslide event, converted into a Landslide Score (0–100). In parallel, the features of the flood branch feed a second similar model (Flood Score, 0–100). The models are trained separately because their targets (an IFFI event against an EM-DAT event) have different physical drivers and different data granularity (Section 1.3.3).
6	Combination and output	Final score = the maximum of the Landslide Score and the Flood Score, a prudential criterion that avoids diluting a high risk concentrated on one front. Output: the score with a colour band, the breakdown of the dominant factors (SHAP), a confidence indicator (lower for the flood branch, given the asymmetric granularity of the historical data), and an exportable report.

A note on **geocoding**, which is the most delicate choice in the pipeline. The public Nominatim/OpenStreetMap instance is fine for prototyping, but its Usage Policy limits use to one request per second, forbids bulk geocoding and does not allow commercial use [30]. Since Phase 1 runs at every query and the Pro and Enterprise plans involve thousands of calls, it cannot be used in production. Two options remain: a self-hosted Nominatim instance, with a non-zero infrastructure cost, or a commercial geocoder. We assume the second, at the unit cost of Section 3.2.3. There is also an accuracy side. In the rural areas and hamlets of Tuscany the irregular house numbering produces positioning errors of 80–150 metres, on a pipeline that computes distances from the river network in the order of a few tens of metres. An error here is not visible in the output, because the pipeline returns a perfectly normal score for the wrong building. This is why Chapter 6 plans a cross-validation with two engines and a spatial confidence indicator.

3.2.2 Data Assets, Licensing and Defensibility

Section 1.5 states that the basic technical mechanism is not defensible: anyone with access to the same public sources can replicate a spatial join plus a gradient boosting. Defensibility lies elsewhere, on three layers.

- **Layer 1, public sources (replicable).** PAI hazard, IFFI, TINITALY, OSM, EM-DAT. Their value is that they are free, not that they are exclusive.
- **Layer 2, derived layer (expensive to replicate, but replicable).** The normalised PostGIS mirror: the seven Basin Authorities publish heterogeneous classifications and

WFS services, and bringing them back to a single schema is work to be redone district by district. Nothing protects it except its own cost, €3,000–5,000 per district (Section 3.4). But that cost is real, and it is the same barrier that makes it hard for a foreign player to come down into the Italian market.

- **Layer 3, proprietary dataset (not replicable).** The archive of assessments compared with real outcomes: paid claims, closed loss assessments, confirmations or refutations of the score. No competitor gets this layer from public sources, and it comes only from a commercial relationship. This is why the 50% discount to the design partner is not a concession: Section 1.5 assumes that public historical events are an acceptable proxy of the claims paid, and the design partner is the only party able to turn that assumption into verified data. The €12,000 of discount are the price of that information.

There is a **licence constraint** on Layer 2 that we must declare. The IFFI inventory and the IdroGEO platform are under CC-BY-SA 4.0 [18], and OpenStreetMap is under ODbL 1.0 [31]: both ask for attribution and add a share-alike clause on derived databases. The single generated report is not a problem; a database that includes those data can be. We do not solve the question here, we state it, and we indicate as a mitigation the architectural separation between the layers of public origin and the proprietary layer of claims and loss assessments, kept as separate databases and not merged, the same kind of regulatory awareness that Section 1.5 applies to the AI Act.

3.2.3 Cost per Assessment and Plan Differentiation

In conversational AI products the cost per session is dominated by usage-based APIs. GeoRisk has an almost zero variable cost: the model is a gradient boosting on tabular data, it does not need a GPU, and inference on a few dozen features is negligible.

Component	**Estimated cost**
Geocoding of the address (commercial service, Section 3.2.1)	€0.005
WFS queries to ISPRA / Overpass / DEM (free; compute and egress cost)	€0.003
Model inference and SHAP computation (no GPU)	€0.0005
Report generation and storage	€0.002
Total per assessment	≈ **€0.01**

Table 3.3: Estimated variable cost of a single assessment.

Against a price of €1.98 per report on the Starter plan (€99 ÷ 50), the contribution margin is **99.5%**: this is the most distinctive fact of the whole model. The plan differentiation follows from it. Since the marginal cost is negligible, the volume limits are not cost constraints but pricing choices: the 50 monthly assessments of the Starter plan align the fee with the value received, they do not reflect a technical threshold. What really tells the plans apart are the features whose cost is fixed: API access, report customisation, service levels, methodological documentation. This is why the Enterprise plan costs more per assessment, and not less.

3.3 Assumptions and Interdependencies

Some assumptions are inherited from Section 1.5. These concern price, cost and structure.

Table 3.4: Business assumptions and where each one is tested.

#	Assumption	Assumed value	Where it is tested
B1	Willingness to pay at list price level	€99 / €399 per month	Survey (Section 5.2)
B2	Share of firms above 5 assessments per month, the threshold below which a subscription is not worth it	To be measured	Survey (Section 5.2)
B3	An SNCE insurer accepts the Enterprise fee	€24,000 per year	Pilot with the design partner (Section 5.2)
B4	Marginal cost per assessment	≈ €0.01	Measurable on the MVP
B5	Data integration cost for a new river basin district	€3,000–5,000 one-off	To be verified on the first district integrated
B6	The design partner grants access to anonymised claims data in exchange for the discount	–	Pilot (Section 5.2)
B7	The architectural separation between the OSM layer and the proprietary layer avoids the extension of the ODbL clause	–	Legal check, not empirical
B8	Reduction of the time to collect and cross-check data compared with the manual process	To be quantified	MVP (Section 5.2)

B8 needs care. Section 1.3.1 correctly says that the computation goes from hours to a few seconds, but the time of a case also includes reading, professional judgement and writing, which GeoRisk does not remove. The overall saving is a hypothesis to measure on the MVP, not a fact.

Four **interdependencies** connect blocks that look independent:

1. **Price ↔ volume threshold.** The Starter fee sets the threshold of five monthly assessments: price and threshold are the same lever, not two.
2. **Data cost ↔ geographic scope.** The cost is territorial and not volume-based, so choosing which districts to cover also chooses the market we can serve.
3. **Explainability ↔ insurer segment.** Without the SHAP layer the product could not be documented for the ORSA: the technical choice of Section 1.3.3 is what makes the Enterprise revenue stream exist.
4. **Dependency on the sources ↔ PostGIS mirror.** The mirror answers the one-way dependency declared among the Key Partners: an operational continuity risk, not a performance optimisation.

3.4 Feasibility and Scalability Logic

The algorithm and the architecture scale at a marginal cost close to zero. Once the pipeline is developed, serving a thousand customers costs as much as serving ten: inference does not need a GPU, the sources are already in a local mirror, and the cost per assessment stays in the order of one cent. The margin grows towards 99.5% as revenue absorbs the fixed costs.

Data coverage does not scale at all. Every new river basin district requires us to integrate and validate the WFS services of the relevant Basin Authority, each with its own classifications and update cycles: €3,000–5,000 one-off per district, plus recurring maintenance. Tuscany falls almost entirely in the Northern Apennines District, which is shared with Liguria and Umbria, so the work done for Tuscany covers only a portion of that district. Six regional areas are still to be covered, which at that unit cost gives €18,000–30,000: a cost per integrated regional area, not per whole district. Adding the pipeline adaptation and the first year of maintenance of the additional connectors, national expansion is in the order of €30,000–50,000 one-off, after which the marginal cost goes back close to zero. One technical constraint stays open: the Italian territory crosses two UTM zones, while the pipeline today uses only one, so expansion to the South also requires a change in the code, and not only the addition of datasets.

The distinction is also useful in strategic terms: if data ingestion were trivial and free, the defensibility claimed in Chapter 2 would collapse, because anyone could replicate the coverage in an afternoon. It also explains why the failure scenario of Section 6.2.6 and its mitigation through PostGIS mirroring make sense.

Economic implication. With a margin close to 100% and costs that are almost all fixed, break-even is not a unit economics problem but a customer counting problem: no volume of use can compromise profitability, and no reduction of the cost per assessment can move it appreciably. The only lever is the number of signed contracts, and the risk is concentrated on adoption, not on technical efficiency. On this basis Chapter 4 builds the market sizing and the break-even.

3.5 Suggested Pricing

Table 3.5: Suggested price list and rationale of each plan.

Plan	Price	Actual cost	Rationale
Free trial	€0	∼€0.05 (5 assessments)	Negligible cost. It serves to overcome the distrust towards an automatic score in a field that is assessed by hand today
Starter	€99/month (50 assessments)	∼€0.50/month	€1.98 per report, 99.5% margin. It is worth it for the customer above 5 assessments per month, compared with pay-per-report
Pro	€399/month (300 assessments)	∼€3/month	€1.33 per report: a real but modest volume discount, because the added value of the plan is API access, not capacity
Pay-per-report	€15–25 per assessment	∼€0.01	No fee. Entry door and coverage of the low volume band
Enterprise	€24,000/year	∼€100/year (10,000 assessments)	Not a volume price: it pays for the API with SLA, support and methodological documentation for the ORSA
Design partner	€12,000 (Year 1 only)	as Enterprise	50% discount as consideration for access to claims data (Section 3.2.2)

The price list is built on perceived value, not on cost: with a marginal cost of €0.01 any cost-plus pricing would produce meaningless figures. The reference is the alternative cost for the customer, the hours needed to collect the same information manually from fragmented sources, and the level of the enterprise solutions, about one order of magnitude above.

4. Addressable Market Analysis

With the prices (Section 3.5) and the costs (Section 3.2.3) of Chapter 3 we answer two questions: how much the market we can really serve is worth, and how many customers we need for the company to cover its own costs. Since one assessment costs about €0.01, the answer does not depend on technical efficiency (Section 3.4). This chapter is therefore, above all, a customer count.

4.1 Market Size Components

- **TAM**: all the Italian firms that could use GeoRisk: intermediaries, loss adjusters and small and mid-sized non-life insurers.

  $$\text{TAM} = (\text{intermediaries in Italy} \times \text{ARPU}) + (\text{target insurers in Italy} \times \text{Enterprise fee})$$

- **SAM**: the part we can serve with the initial infrastructure: only the Central-North, and only firms above the volume threshold where a subscription is worth it.

  $$\text{SAM} = (\text{Central-North intermediaries} \times \text{share above threshold} \times \text{ARPU}) + (\text{Central-North insurers} \times \text{Enterprise fee})$$

- **SOM**: the customers estimated at the end of the third year.

  $$\text{SOM} = (\text{paying firms in Year 3} \times \text{ARPU}) + (\text{insurers under contract in Year 3} \times \text{Enterprise fee})$$

As stated in Section 3.1, the two revenue streams follow different logics and we compute them separately. In the **Intermediaries** stream the unit is the annual subscription of an adjusting firm or a broker, and the ARPU (Average Revenue Per User) is how much a customer pays on average in one year. In the **Insurers** stream the unit is the annual Enterprise licence.

We declare two exclusions, so that we do not inflate the calculation. **Large national and international insurers**, for the reason already given in Section 3.1. And **banks**: they value properties as collateral for mortgages and could in principle use a risk score, but serving them would mean integrating GeoRisk into their Automated Valuation Models, and so building a different product from the one of Chapter 3.

4.2 Estimation Methodology

The **top-down** method starts from a large industry figure and applies a market share to it. The **bottom-up** method starts from the concrete units of the business and adds them up. We discard the top-down: starting from the cat-nat insurance gap (€1.7 billion of premiums not collected, Section 1.1.3) would give huge but meaningless figures, because that gap is made of insurance premiums, not of software licences, and any linking percentage would be invented. The bottom-up starts from verifiable numbers, the official registers of intermediaries and insurers, and from the price list we have already decided. Every assumption stays visible and can be tested one at a time with the methods of Chapter 5.

4.2.1 Scope and Filters

Starting population. The IVASS Registro Unico degli Intermediari counts 30,942 members in Sections A and B, agents and brokers [32]. They are not all potential customers, because many work almost only on motor or life insurance, where the landslide or flood risk of a building does not enter the assessment. The register is not broken down by portfolio type, so we apply a declared filter: we assume that 60% have a non-motor non-life portfolio focused on businesses: **18,565 firms**. We add the loss adjusters who work on preliminary estimates and catastrophe claims: AIPAI represents about 300 firms and more than 2,000 specialised loss adjusters [29] and, counting those who are not members, we estimate **3,500 firms**. Starting base: **22,065 firms**. On the insurer side, IVASS supervises about 90 companies based in Italy. Removing the large groups (Section 3.1) and those that operate only in life insurance, about **45 small and mid-sized non-life insurers** with a regional property portfolio remain.

First filter: geography. We start from the Central-North for two reasons. The technical one is that every integrated regional area costs €3,000–5,000 of data engineering (Section 3.4), and expansion to the South also requires a change in the pipeline, which today works on a single UTM zone. The market reason is that the Central-North has the highest industrial density, and therefore more businesses subject to the cat-nat duty. We assume that it holds 60% of the intermediaries (13,239 firms) and 27 of the 45 target insurers.

Second filter: volume threshold. It comes from Section 3.1: with the Starter plan at €99 per month and pay-per-report at about €20, a subscription is worth it only above five monthly assessments. Below that, the rational choice is to buy single reports, and a firm that buys two per year is not a recurring customer. In the base case we assume that **40%** of the firms go above five monthly assessments, which is assumption B2 of Section 3.3 and the most fragile one in this chapter, because it does not come from public data but from our own estimate. This is why it is the first thing that the survey of Section 5.2 measures, and the failure threshold set in Section 5.3 (less than 20% of the respondents) matches the pessimistic scenario of Section 4.2.4.

Applying the two filters: **13,239 $\times$ 40% = 5,296 firms in the Central-North that can be converted to a subscription**.

Check from the demand side. About 6.1 million businesses are registered in Italy [36]. Those that own properties or capital assets to insure and that fall under the duty of L. 213/2023 are in the order of 1.2 million. If each of them needs at least one assessment when the policy is renewed, national demand is above one million assessments per year, about 720,000 in the Central-North. On the supply side, 5,296 firms with ten assessments per month on average produce about 636,000 assessments per year. The two figures are of the same order, which makes the 40% assumption at least plausible.

The ISPRA dataset on businesses exposed to hydrogeological instability [37] confirms the same order of magnitude in the pilot area: in Tuscany 102,517 businesses out of 366,692 fall in areas with medium hydraulic hazard (28.0%) and 9,078 in areas with high or very high landslide hazard (2.5%), reported separately because the two classifications use different scales and partly overlap.

We must state one limit: the fact that demand exists does not prove that it passes through subscribed adjusting firms, because part of those assessments will be made directly by the mandating insurers, the risk described in Section 6.2.2.

4.2.2 Revenue per Customer and Launch Funnel

Intermediary ARPU. The price list of Section 3.5 has the Starter plan at €99 per month (€1,188 per year) and the Pro plan at €399 per month (€4,788 per year): the Starter plan for the small firm, the Pro plan for the structured broker who uses the API. Assuming 85% Starter and 15% Pro:

$$(0.85 \times €1{,}188) + (0.15 \times €4{,}788) = €1{,}009.80 + €718.20 = €1{,}728 \text{ per year}$$

We round prudentially to **€1,700 of yearly ARPU**, without counting the pay-per-report revenue from subscribers who go over their plan. That revenue exists, but it is hard to estimate, and leaving it out makes the calculation more solid.

Revenue per insurer. The Enterprise fee is €24,000 per year (Section 3.5). The design partner of the first year pays €12,000. As explained in Section 3.2.2, the discount is the price of the claims data, so it must be read as a cost of acquiring that data.

First year funnel (Tuscany). Tuscany holds about 6.4% of the Italian firms (1,412), of which 565 are above the volume threshold. In the first year we do not reach the whole pool: with the association channels and the trade fairs of Section 3.1 we assume that we contact 20% of the Tuscan firms (282), with a conversion to paying customer of 4%: **11 subscribers** at the end of the year. These numbers are deliberately small, as a methodological choice: 4% out of 282 contacts is exactly what the landing page test of Section 5.2 can confirm or disprove in a few weeks, and the success threshold of Section 5.3 ("at least 11 out of 282") is the same assumption used here.

4.2.3 Market Sizing and Three-Year Plan

Level	Intermediaries	Intermediary revenue	Insurers	Insurer revenue	Total
TAM (Italy)	22,065 firms	~€37.5 M	45	~€1.1 M	~€38.6 M/year
SAM (Central-North, above threshold)	5,296 firms	~€9.0 M	27	~€0.65 M	~€9.6 M/year
SOM (Year 3)	165 paying firms	~€280,500	4	€96,000	~€377,000 ARR

Table 4.1: TAM, SAM and SOM by revenue stream.

The SOM of the third year is worth 3.9% of the SAM: 3.1% of the firms in the Central-North that can be converted to a subscription, and 14.8% of the target insurers in the area.

Metric / Year	Year 1 – Tuscany pilot	Year 2 – Central-North	Year 3 – Consolidation
Firms contacted (cumulative)	282	∼1,900	∼3,400
Paying firms at year end	11	60	165
Insurers under contract	1 (design partner)	2	4
Assumed monthly churn	5%	4%	3%
Intermediary revenue	€18,700	€102,000	€280,500
Insurer revenue	€12,000	€48,000	€96,000
Total revenue (ARR at year end)	**€30,700**	**€150,000**	**€376,500**

Table 4.2: Three-year plan by segment.

Churn, that is the share of customers who cancel the subscription every month, goes down for a structural reason and not out of optimism. In the first year the product covers only one district and the archive of assessments compared with real outcomes is almost empty. From the second year the proprietary dataset of Layer 3 (Section 3.2.2) starts to build up, the score becomes more reliable and changing tool costs the customer more. The 3% of the third year stays well below the 7% alert threshold of Section 6.3.

Already in the second year the insurers generate almost a third of the revenue, even though they are 2 customers out of 62. That confirms in numbers what Section 3.1 says: intermediaries serve to validate the model and to build the dataset, while it is the Enterprise contracts that pay the fixed costs.

4.2.4 Sensitivity Analysis

The two most fragile assumptions are the share of firms above the volume threshold and the conversion rate of the first year. We vary them together because in reality they move in the same direction, since a firm with few assessments is also less likely to subscribe.

In the **pessimistic scenario** the SAM falls to about €5.0 M, a little more than half the base case. Section 5.3 defines exactly this condition as a failure signal for the survey: with a SOM of €190,000 in the third year it would not cover the fixed costs (€227,000). The answer is not to wait, but the pivot already planned in Section 6.3: give up selling to small firms and to focus on API contracts with aggregated firms and on the Enterprise channel, which is worth twenty times more per customer and does not depend on the five assessment threshold. The **optimistic scenario** does not change the nature of the problem: even at €12.9 M the SAM stays a niche. A small SAM is a useful feature here, because it is exactly the size of the market that slows down the entry of competitors (Section 4.3.3).

4.3 Investment Decision Support

4.3.1 Cost Structure and Break-Even

With a variable cost of about €0.01 per assessment the contribution margin is 99.5%. In classic Cost-Volume-Profit analysis break-even depends on the ratio between price and variable cost, but here the variable cost is so close to zero that break-even almost exactly matches the fixed costs. No technical optimisation can move it, and the only lever is the number of contracts.

Fixed costs (bottom-up on the items of Section 3.1)	Year 1	Year 2	Year 3
Technical team (GIS + ML, 2 FTE at launch)	€90,000	€110,000	€130,000
Cloud, PostGIS database, legal advice and overheads	€30,000	€34,000	€40,000
Ingestion of new districts and connector maintenance	€10,000	€18,000	€22,000
Marketing and association channels	€25,000	€30,000	€35,000
Total fixed costs	**€155,000**	**€192,000**	**€227,000**

Table 4.3: Fixed cost structure over the first three years.

The third row is the accounting translation of Section 3.4: it grows with the number of river basin districts covered, not with the number of customers, and connector maintenance keeps growing even without acquiring a single new customer, because every Basin Authority updates its layers with its own timing and schemas.

Summary income statement	Year 1	Year 2	Year 3
Revenue	€30,700	€150,000	€376,500
Variable costs	~€150	~€300	~€600
Fixed costs	€155,000	€192,000	€227,000
Result	− **€124,450**	− **€42,300**	+ **€148,900**

Table 4.4: Summary income statement over the first three years.

The break-even revenue of the third year is €227,000 ÷ 0.995 ≈ €228,000 of yearly recurring revenue. Starting from the €150,000 at the end of the second year, and growing at the planned rate, break-even arrives around **month 28**. Translated into customers, the €228,000 can be reached in three ways:

- with intermediaries only: **134 firms**;
- with insurers only: **10 Enterprise licences**;
- with the mix planned here: **4 insurers** (€96,000) plus **78 firms** (€132,600).

The three paths do not carry the same risk. 134 firms require an association channel that works continuously for two years; 10 insurers require ten long sales cycles, each one with the authorisation of the Compliance function described in Section 5.1. The intermediate mix is the only realistic one, and it is the reason why Section 3.1 keeps both segments: neither of them alone reaches break-even in three years.

4.3.2 LTV, CAC and Capital Requirements

LTV (Lifetime Value) is the margin a customer brings over their whole useful life, and **CAC** (Customer Acquisition Cost) is how much we spend to acquire them. The ratio should stay above 3x.

Intermediaries. With a monthly churn of 3% a subscriber stays on average 1 ÷ 0.03 ≈ 33 months, about 2.75 years.

- LTV = €1,700 × 0.995 × 2.75 ≈ **€4,650**
- CAC = €30,000 of second-year marketing budget ÷ about 52 new firms acquired, plus a share of sales time ≈ **€600**
- LTV/CAC ratio = **7.8x**, with CAC payback in a little more than four months

Even in a stress scenario, with churn at 5% per month and CAC at €900, the ratio stays at 3.1x.

Insurers. The licence is worth €24,000 per year at almost full margin.

- LTV ≈ **€95,000**, over an average duration of four years, justified by the cost of replacing an API integration already part of the underwriting processes
- CAC ≈ **€8,000**, between sales time, pilot support and the production of the methodological documentation for the ORSA
- LTV/CAC ratio ≈ **12x**

Here the real constraint is not the cost but the length of the sales cycle: a supervised insurer buys after a pilot and after the authorisation of Compliance, and a favourable CAC does not shorten that path.

Capital requirement.

- MVP development before launch: about €70,000, six months of technical work with no revenue
- Losses of the first two years: €124,450 + €42,300 = €166,750
- Peak cash deficit, at the end of the second year: about **€236,750**

We set the initial capital at **€250,000**, with a safety margin. This figure is before risks, and Section 6.3 will revise it to €280,000 once the cost of the safeguards is put in the budget. The third year closes with a profit of €148,900, but this is not enough to return all the capital invested: at the end of the third year the cumulative balance is still negative by about €87,850, and full recovery happens during the fourth year, consistently with the deliberately prudent conversion rates of Section 4.2.2. The most effective lever to bring it forward is the Enterprise channel: four more licences are worth €96,000 at almost full margin, they would cover the remaining deficit on their own and they would bring payback forward by about one year.

4.3.3 Strategic Implications

Four conclusions, from transparent calculations on assumptions that are declared and that can be verified with the methods of Chapter 5.

1. The niche is itself a defence. A SAM of €9.6 M does not justify the entry of an international enterprise player such as Verisk or Cytora, which would have to bear the data engineering cost of seven Italian river basin districts (Section 3.4) for modest revenue. The most likely competitive risk comes from the Italian business information incumbents, CRIF and Cerved, which have already borne those commercial relationship costs: this is the scenario of Section 6.2.3. Eoliann and SaferPlaces also remain relevant (Chapter 2), but today they cover only one peril and they address customers different from ours.

2. The loss of the first two years is structural, not an estimation error. With costs that are almost all fixed and a margin of 99.5%, the company loses money until it reaches the break-even volume, and from there almost every extra euro becomes margin: the curve stays flat and then rises sharply. The goal of the first year is not profit but the proprietary dataset of Layer 3 (Section 3.2.2) and the validation of the assumptions of Chapter 5.

3. Break-even depends on the mix, not on the volume. An Enterprise contract of €24,000 is worth as much as fourteen Starter subscriptions. In the third year revenue grows by about €18,900 per month, so one more insurer brings break-even forward by about a month and a half, while the big effect is on payback, where four additional contracts are worth a whole year. An operational priority follows: the pilot with the design partner (Section 5.2) does not only serve to collect claims data, it is the first step of the revenue stream that supports the fixed costs.

4. The bottleneck is the channel, not the demand. The duty of L. 213/2023 generates more than a million potential assessments per year, much more than any figure used here. The risk is not that the need is missing, but that it is met elsewhere, by the rough estimates of the mandating insurers or by a free module included in agency management software. This is why the two most exposed assumptions, the threshold of five monthly assessments and the 4% conversion, are the first ones that Chapter 5 puts to the test, with the cheapest experiments.

Overall the analysis supports the decision to invest, but under precise conditions: about €250,000 of initial capital (€280,000 in the risk-adjusted version of Section 6.3), two years of expected loss, break-even around month 28, and return on the investment in the fourth year. The project holds if the Enterprise channel opens within the second year. If it does not, we fall into the pessimistic scenario of Section 4.2.4, which Section 6.2.1 then examines as a failure cause.

5. Personas and Validation Strategy

Chapter 4 estimates the reachable market starting from explicit assumptions: the share of intermediaries reached by the promotion of the tool, the sign-up rate to the waitlist, the conversion from free trial to paying licence, and continuous use in the pilot with the design partner. Here we define who to observe in order to validate them, the personas, and how to test them with real people, in a cheap and fast way, before we commit significant resources.

5.1 User Personas Definition

The reference market is the B2B side of Italian insurance distribution: the about 30,942 intermediaries listed in the Registro Unico degli Intermediari Assicurativi (RUI) kept by IVASS, the insurance supervisory authority (Sections A and B of the register, agents and brokers respectively) [32]. There are three personas. The first two are both buyer and user, while the third one authorises the purchase but does not use the product.

Persona	**Role in the purchase**	**In short**
Independent loss adjuster / broker	Buyer and user	Assesses the risk of a building on behalf of an insurer. Today they do it by hand, with maps that are too coarse and with scattered sources.
Underwriter of a regional insurer	Buyer and user	Decides the premium of a policy. Today they are forced to set the same prices for different risks, because they do not have precise enough data.
Head of Compliance and Risk Management	Buyer, not user	Authorises the purchase in order to show that the insurer complies with a new EU regulation on climate risk.

Table 5.1: The three user personas of GeoRisk.

Independent loss adjuster / broker. Listed in the RUI (Section A or B), owner or employee of a small firm, they assess the risk of a specific building for one or more insurers. They work with the tools described in Section 1.2: the hazard maps of the Piano di Assetto Idrogeologico (PAI, the official instrument that classifies the territory by hydraulic and landslide risk), which are binary and at municipal scale, plus scattered sources that do not scale to the volumes of the new insurance duty for businesses against natural disasters ("cat-nat", 2024 Budget Law). *Goal*: to get quickly a risk score specific to the building, which they can justify to the mandating insurer.

Underwriter of a regional insurer. In the underwriting team of a small or mid-sized insurer, they decide whether to insure a building and at which premium. They do not have the enterprise scoring tools (Verisk, Swiss Re/Fathom, the systems used by the large international companies), which are too expensive, and they are forced to apply undifferentiated premiums to buildings with different exposure in the same area. *Goal*:

to differentiate the premium according to the real risk of the building, without exposing the insurer to losses that adequate premiums do not cover.

Head of Compliance and Risk Management. Responsible for the ORSA assessment (Own Risk and Solvency Assessment, the periodic risk self-assessment that every insurer must produce by law) in a small insurer with SNCE status (Small and Non-Complex Enterprise, a status that the regulation reserves to insurers below a certain size threshold, with simplified obligations). They decide whether to authorise the adoption of the tool. Directive (EU) 2025/2 (art. 45a) requires climate risk scenarios in the ORSA assessment, but the insurer has no budget for enterprise solutions. *Goal*: to show regulatory compliance with a solution that is proportionate to the size of the insurer.

5.2 Validation Strategy Methods

The path has an increasing cost: each phase starts only if the previous one gave enough feedback.

At the time of writing we do not have a direct contact with a loss adjuster or a broker for a structured interview. We replace this first step with the analysis of two recent and verifiable secondary sources: the CINEAS white paper (Politecnico di Milano, 2023) [15] and the public statements of AIPAI, the Italian association of insurance loss adjusters [33, 34, 35]. If a real contact came up, the direct interview should be added as a priority step, before the survey.

Table 5.2: Validation methods and the assumption each one tests.

Method	**What we do**	**Assumption tested**
1. Analysis of secondary sources	Analysis of the CINEAS white paper and of the public statements of AIPAI on the challenge that the cat-nat duty represents for loss adjusters as a professional category.	The problem is recognised by the sector, and not only deduced from ANIA and ISPRA data.
2. Online survey to loss adjusters and brokers	A short survey distributed through professional groups and trade associations, addressed to a sample of independent loss adjusters and brokers in Tuscany.	Assumptions B1 and B2 of Section 3.3: that the firms go above five monthly assessments and that they accept the price list of Section 3.5 (Starter €99/month, Pro €399/month).
3. Pre-launch landing page	A page that describes the tool (flood and landslide, Tuscany area), collecting contacts for a waitlist.	The willingness to leave a real contact for a tool that does not exist yet.
4. MVP	A prototype limited to the pilot area (Tuscany) and to the flood and landslide perils: a risk score per building, computed by crossing the ISPRA data (hydraulic and landslide) with the building geometries, released to the people on the waitlist.	The score is perceived as more useful and faster than checking the PAI maps by hand.
5. Pilot with a design partner	Use of the MVP for a few weeks with an adjusting firm or a small regional insurer in Tuscany, under an informal agreement.	Continuous use leads to faster assessments and to more differentiated premiums than the manual process.

5.3 Learning and Decision-Making

Every phase has a failure threshold and a success threshold, both defined before the test starts.

Table 5.3: Failure and success signals defined before each test.

Test	Failure signal	Success signal
1. Secondary sources	No source confirms the problem as relevant for the sector.	The sources explicitly confirm that the current tools are inadequate (for example CINEAS: 75% of homes exposed to significant risk, less than 5% with adequate cover).
2. Survey	Less than 20% of the respondents go above five monthly assessments, the pessimistic scenario of Section 4.2.4 that reduces the SAM to €5 M, or the majority judges the Starter fee out of reach.	At least 40% of the respondents report more than five assessments per month (assumption B2, base case of Section 4.2.1) and at least one third judges the Starter fee of €99/month acceptable.
3. Landing page	Less than 2% of the qualified visitors leave a contact: half the penetration on which the first year plan is based.	At least 4% leave a contact, 11 out of 282 firms reached, the same penetration assumed for Year 1 in Section 4.2.2.
4. MVP	Less than one third of the people on the waitlist run more than one assessment in the first four weeks.	At least half of them come back to use the tool, with at least five assessments per active user in the first month: the volume that makes a subscription rational (Section 3.1).
5. Pilot	The design partner does not provide comparable real outcomes, and assumption B6 of Section 3.3 stays unverified.	The design partner provides at least 50 real outcomes (closed loss assessments or settled claims) to compare with the assigned scores: the first core of Layer 3 of Section 3.2.2, the only defensible barrier of the project.

Three rules guide the decisions. **Persevere**: if a phase gives a success signal we move to the next one, and the observed value replaces the corresponding assumption in the model of Chapter 4 (for example the real conversion rate of the landing page replaces the initial adoption assumption). **Pivot**: if the problem is confirmed but the solution does not convince (a positive survey but an MVP that is little used), we change one specific element, such as the pricing, the report format or the pilot area, without giving up the project, and we repeat only the cheapest test that is useful to check the change. **Stop**: if the secondary sources and the survey do not confirm that the sector feels the problem, or if the pilot shows no improvement in premium differentiation, the basic assumption does not hold. Defining these conditions in advance makes the validation honest: we state from the start which evidence would show that we are wrong.

6. Failure Analysis and Risk Mitigation

6.1 Negative Backcasting Method

So far we have explained why GeoRisk should work. Now we start from the idea that it has already failed. We place ourselves at month 36, the company has closed, and we do not ask whether things could have gone wrong but **exactly how** they went wrong. This is **Negative Backcasting**: we start from the result and go back to the causes. The method is useful because the question "will this risk happen?" almost always gets a "probably not", while "since it happened, what caused it?" forces us into the details.

Chapter 4 already points at the fragile spot. With a margin of 99.5%, technology cannot make the company fail for economic reasons, and that chapter closes with a precise condition: the project holds if the Enterprise channel opens within the second year. Here we explore the other side of that sentence with six scenarios, ordered not by severity but by how likely they are to be the real cause of the shutdown.

Scenario	**Probability**	**Impact**	**When it hits**
Section 6.2.1: The Enterprise Channel Does Not Open	High	Fatal	Months 12–24
Section 6.2.2: Small Firms Do Not Subscribe	High	Fatal	Months 6–18
Section 6.2.3: Incumbents Give the Feature Away for Free	Medium	Fatal	Months 24–36
Section 6.2.4: Geocoding Error and Lawsuit	Low	Fatal	Any moment
Section 6.2.5: Climate Drift of the Model	Low as a direct cause	Indirect but cumulative	Months 18–36
Section 6.2.6: Regulatory Shock on Data and the AI Act	Low	Severe	Months 12–24

Table 6.1: The six failure scenarios, ordered by how likely they are to be the real cause of the shutdown.

Climate drift is in fifth position because, unlike the others, it is not a single event but a process that builds up: it rarely kills on its own, but it silently weakens all the others.

The fact that the first two scenarios are commercial and not technical does not devalue the technical work. It follows from the cost structure of Section 4.3.1: an algorithm that works very well and is not bought produces the same result as one that does not work. We must add a risk that is not a scenario but a permanent condition: the team is two people, and if one of them leaves in the first eighteen months the project stops, whatever scenario is running. At this scale there is no structural mitigation.

Almost all six are classified as fatal for one reason. At month 24 the plan reaches its peak cash deficit, €236,750 against a capital of €250,000 (Section 4.3.2, revised in Section 6.3): a buffer of €13,250, about 5%, that any one of these scenarios consumes on its own. Failure is not an event: it is a margin that gets thinner month after month until it disappears.

6.2 Common Failure Causes

6.2.1 The Enterprise Channel Does Not Open

The plan of Section 4.2.3 has four insurers under contract in the third year, €96,000 of revenue, a quarter of the total. In Section 4.3.1 it is the condition without which break-even would need 134 firms instead of 78.

The first contract always arrives, because the design partner pays €12,000 in exchange for the data (Section 3.2.2) and it is a small risk for them. The second one does not arrive, and the reason is not the price but the time. As described in Section 5.1, a supervised insurer buys after a technical pilot and after the authorisation of Compliance, and every cycle lasts between nine and fifteen months. A cycle started in month 10 closes, at best, in month 22. In the meantime the design partner reaches the end of the discount and proposes to stay at €12,000 instead of moving to €24,000. Accepting halves the only revenue stream that covers the fixed costs, and refusing loses it completely. Without the second insurer the revenue of the second year falls from €150,000 to about €114,000, the loss grows from €42,300 to about €78,000, and the peak deficit goes beyond the capital even before month 24.

The hardest signal to recognise is the absence of a signal: in month 12 no second insurer has reached the technical pilot yet. Not a refusal, just silence.

6.2.2 Small Firms Do Not Subscribe

Assumption B2 of Section 3.3, 40% of the firms above five assessments per month, turns out to be optimistic: the survey of Section 5.2 records about 20%, the pessimistic scenario of Section 4.2.4 that reduces the SAM to €5 M.

The need is not missing: Section 4.2.1 verified that there are more than one million assessments per year. What is missing is the passage through subscribed firms, because the loss adjuster receives a free rough estimate from the mandating insurer that is enough for most cases, or checks the regional portal in twenty minutes without paying anything. For someone who does two assessments per month a fee of €1,188 per year is simply more expensive than the alternative, and the users who register stay on occasional pay-per-report. The behaviour is exactly the one Section 3.1 predicts, but it does not generate the recurring revenue that the fixed costs of Section 4.3.1 need.

With 60 firms in the third year instead of 165, revenue from intermediaries falls from €280,500 to about €102,000 and break-even moves beyond the fourth year. Chapter 5 can catch this scenario, and only this one, before we spend the €70,000 of the MVP, with a landing page and a survey that cost a few weeks, which is why Section 5.3 puts them first.

6.2.3 Incumbents Give the Feature Away for Free

The niche defence of Section 4.3.3 holds against Verisk, but not against those who are already inside Italy. CRIF and Cerved already sell commercial information to the same insurers and to the same intermediaries: adding a Cat-Nat score to the catalogue only requires one more line on an existing contract. The most silent threat, however, comes from the software houses of agency management systems: if the Cat-Nat module appears for free in the program that the broker opens every morning, it does not have to be better than ours in order to be used instead of ours.

The only defence identified in Section 3.2.2 is Layer 3, the archive of real claims data. In month 24, however, it holds only a few dozen cases, because it depends on the design partners of the scenario of Section 6.2.1 that did not materialise. The barrier was not knocked down, it was never built. The product becomes indistinguishable from an included module, negotiations stop and churn speeds up, because the customer now has a zero cost alternative.

6.2.4 Geocoding Error and Lawsuit

To keep initial costs down the platform uses a single geocoding engine, that is the step that turns a written address into coordinates on the map: Nominatim on OpenStreetMap data, which is free. In the rural and mountain areas of Tuscany house numbering is irregular and the returned point can fall 80–150 metres from the real building. On a map at municipal scale this changes nothing, but it becomes decisive when we overlay it on high resolution data such as the IFFI Inventory, where the boundary of an active landslide polygon can pass a few metres from a building. The granularity asymmetry of Section 1.3.3 comes back here, applied to the input instead of to the sources.

In the concrete case an industrial warehouse is placed 120 metres further downhill than its real position and receives a score of 10 out of 100, while it stood inside an active landslide polygon. After the landslide event the insurer challenges the assessment, the business sues the broker for failure to advise, and the broker seeks recourse against us. The defence costs exceed the remaining capital of a startup in its second year, but the worst blow is not economic: in a small professional category, where everybody knows everybody, a report that is demonstrably wrong circulates in a few weeks. We must point out that Section 4.3.1 has no item for professional civil liability: a real gap in the financial plan, which Section 6.3 corrects.

6.2.5 Climate Drift of the Model

The model trains on historical events within a window of 15–30 years (Sections 1.3.3 and 1.5), under the implicit assumption that the risk observed in the past stays representative of the present. The assumption is reasonable for a stable phenomenon, but not for one that by definition is changing, because the frequency of extreme events grows, and Section 3.1 already pointed at this as the reason for periodic retraining. The problem is what happens if that retraining stays generic, without an explicit check on how far the model has moved away from reality.

Unlike the geocoding error, here there is no wrong input to correct: the model keeps answering and the scores stay plausible, but a building whose real exposure has grown receives a score computed on a risk that no longer exists, systematically lower than the true one. No single report is “wrong” in the way a badly geolocated address is: it is the whole set that slides. The point where it becomes visible is Layer 3, the archive that compares scores with real outcomes. If calibration drifts, we see it there, with claims occurring in areas rated as low risk more often than the estimated probability predicted. The archive that is our only defensible asset is therefore also our only instrument for detecting this failure.

It is not fatal in itself, but it wears down the trust of the design partner exactly when that relationship is the most important competitive barrier. And if it stays unchecked for a long time the risk gets worse, until it looks like the previous scenario, because an underrated building that suffers real damage exposes us to the same legal liability.

6.2.6 Regulatory Shock: Data and the AI Act

The initial architecture takes for granted that the WFS/GeoJSON services of ISPRA IdroGEO stay open and can be queried in real time. In month 18 the portal introduces a limit on the number of calls, for load reasons. The portal is not being hostile: for a public service that was never designed for commercial traffic, capping calls is routine administration, but for us it means that the promise of generating reports in real time stops working in a predictable way.

In parallel, Section 1.5 assumes that GeoRisk stays outside the "high risk" category of the AI Act, on a literal reading of Annex III that explicitly names only life and health: a defensible reading, but still a reading. An interpretative communication from the Commission, prompted by consumer associations that challenge the exclusion of non-life insurance, extends the classification to geo-climatic scoring systems that affect the price of a policy. From that moment formal technical documentation, structured human oversight, registration in a European database and periodic audits all apply.

The scenario does not kill on its own, but it eats up the buffer that is not there: a compliance cost of €30,000–50,000 in month 20 closes the company even if everything else goes well. It has low probability and long timescales, and exactly for this reason the right mitigation is not financial but a matter of attention: noticing it twelve months in advance turns it from a shock into a planned expense.

6.3 Early Warning and Mitigation

These thresholds follow those of Section 5.3, with a difference in timing: the ones of Chapter 5 are for before the launch, when the possible answer is still "we do not start", while these are for afterwards, when the possible answer is to correct the course. We set a **quarterly review** where we check all the indicators together, because a signal that is looked at only when we already suspect a problem arrives too late.

Table 6.2: Early warning signals and planned response for each failure cause.

Failure cause	**Early warning signal**	**Mitigation and response**
Enterprise channel (Section 6.2.1)	No second insurer in technical pilot by month 12, or a renewal request below €18,000	Three negotiations in parallel instead of one: with cycles of 9–15 months, a single negotiation is not a channel. ORSA documentation prepared in advance, since Compliance asks for it first. *If it triggers*: third-year ambition down from four insurers to two, or the channel moved to aggregated agency networks, which decide faster.
Firms that do not subscribe (Section 6.2.2)	Churn above 7% per month against the 5% assumed for the first year; or more than 65% of reports sold as single purchases; or LTV/CAC below 3x	No subscription pushed on those below five monthly assessments, where Section 3.1 says it is not worth it for them: annual packages of reports instead. *If it triggers*: the pivot of Section 4.2.4, from single firms to API contracts with a guaranteed annual minimum.

Failure cause	Early warning signal	Mitigation and response
Commoditisation (Section 6.2.3)	A free Cat-Nat module announced in a competing management software, or a Layer 3 archive growing by less than 50 validated cases per quarter	That archive treated as the company's first indicator, before revenue, with AIPAI agreements for validation. Focus on the explainability of the score: the report also says *why*, which a generic included module does not. *If it triggers*: we move where an included module does not reach, complex loss assessments, litigation and ORSA documentation.
Geocoding error (Section 6.2.4)	More than 1.5% of addresses reported as wrong, or even a single formal written complaint from a loss adjuster	Four safeguards: a professional indemnity policy active from the first report issued, not from the first problem; a second geocoding engine, called only when the first returns low confidence, so the cost stays on a fraction of the calls; a spatial confidence score, separate from the analytical confidence of Section 1.3.4, so that below the threshold the report says the address is unreliable instead of returning a number; a clause excluding liability for decisions taken without a site inspection. *If it triggers*: reports in the area suspended for manual check, before answering the customer.
Climate drift (Section 6.2.5)	A systematic deviation between predicted probability and real outcomes in the Layer 3 archive, at any quarterly review; or no retraining in the last twelve months	An explicit calibration check inside the retraining already planned in Section 3.1, not as a separate activity. *If it triggers*: the branch concerned retrained on the most recent data, and the design partner told about the deviation before they find out on their own.
Regulatory shock (Section 6.2.6)	ISPRA API availability below 95% per month; or average latency above 3.5 seconds; or drafts to postpone the Cat-Nat duty; or any Commission communication on Annex III referring to non-life insurance	A periodically updated local copy of the data in PostGIS, which decouples the service from the instant availability of the public portals, plus the technical documentation for "high risk" systems prepared in advance even without an obligation. *If it triggers*: compliance activated on documentation already written: the cost stays, the months of downtime do not.

These measures are not free, and Section 4.3.1 does not include them.

Additional item	Yearly cost
Professional indemnity policy	€3,500
Continuous legal monitoring on the AI Act	€4,000
Second geocoding engine and data mirroring	€4,500
Model calibration check	€1,500
Total	**€13,500**

Table 6.3: Additional yearly cost of the mitigation measures.

The legal monitoring is incremental with respect to the generic legal advice already present in Section 4.3.1, because it covers continuous monitoring and not one-off advice. Applying these items to the plan of Chapter 4:

	Year 1	Year 2	Year 3
Fixed costs of Section 4.3.1	€155,000	€192,000	€227,000
Fixed costs with mitigations	€168,500	€205,500	€240,500
Result	– **€137,950**	– **€55,800**	+ **€135,400**

Table 6.4: Risk-adjusted plan after the mitigation measures.

The break-even revenue rises from €228,000 to about €242,000: eight more subscriptions at the ARPU of €1,700 of Section 4.2.2. The peak cash deficit goes from €236,750 to about €263,750, above the €250,000 planned in Section 4.3.2, so the initial capital must be revised to **€280,000**, with a buffer of 5.8% instead of almost zero. Defending against all the scenarios of this chapter therefore costs about 6% of the fixed costs and moves break-even by eight customers, which is little. The €30,000 of extra capital are a larger figure, but they do not pay for the safeguards: they correct a buffer that was too thin in the original plan.

6.4 Strategic Thinking Benefits

The critical point is commercial, not technical. Of the six scenarios, the three most likely ones concern sales and competition. With only two people, the time spent improving the score is time taken away from opening the Enterprise channel, which is what keeps the company alive. Beyond a certain accuracy, more accuracy does not sell anything.

Technical risk does not always threaten in the same way. The geocoding error has no impact on costs, because at €0.01 per assessment being wrong costs the same as being right, but it exposes us to a lawsuit and to reputational damage in a small professional category. Climate drift is more insidious, because there is no single report to point at as wrong, only a deviation that builds up until someone measures it. In both cases the safeguard is the same: a product that reports low spatial confidence, or a calibration that has drifted, instead of returning a number anyway.

The only competitive barrier depends on the customer we risk not having, and it is also the first place where problems would show. Layer 3 comes from the design partners of the scenario of Section 6.2.1 and it is also where a drift of the model would become visible first: the risks of this chapter are not independent, they converge on the same archive. The pilot of Section 5.2 is therefore not only a technical test: it starts the competitive barrier and the monitoring at the same time.

Knowing in advance what to do is worth more than hoping it will not be needed. Every scenario has an answer decided today and a numerical threshold that says when to activate it. A threshold decided in advance is what allows the change of direction to happen while there is still capital to fund it.

The bottom line. The problem is real and the regulation makes it urgent. The technology works, but it is not defensible on its own, so the bet is commercial before it is technical. The numbers say it is worth trying under the conditions already set out in Section 4.3.3, with the capital revised to €280,000 by this chapter. One condition counts more than the others: the Enterprise channel must open within the second year, because that is where both the revenue that covers the fixed costs and the data that build the only defensible barrier come from. If it does not open, the problem is not the product but the project.

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